

Transformation Type	Expression	Trig Identities	
Shift Right	$f(x - a)$	$A\sin(\alpha) + B\cos(\alpha) = \sqrt{A^2 + B^2} \cos\left(\alpha - \tan^{-1}\left(\frac{A}{B}\right)\right)$ $\cos(\alpha) \cos(\beta) = \frac{1}{2}(\cos(\alpha - \beta) + \cos(\alpha + \beta))$ $\sin(\alpha) \sin(\beta) = \frac{1}{2}(\cos(\alpha - \beta) - \cos(\alpha + \beta))$ $\sin(\alpha) \cos(\beta) = \frac{1}{2}(\sin(\alpha - \beta) + \sin(\alpha + \beta))$	
Shift Left	$f(x + a)$		
Shift Up	$f(x) + a$		
Shift Down	$f(x) - a$		
Vertical Stretch	$af(x), \quad a > 1$		
Vertical Compress	$af(x), \quad 0 < a < 1$		
Horizontal Compress	$f(bx), \quad b > 1$		
Horizontal Stretch	$f(bx), \quad 0 < b < 1$		
Reflect over x-axis	$-f(x)$		
Reflect over y-axis	$f(-x)$		

Polynomial Function	$f(t) = \sum_{k=0}^n a_k t^k$	Fundamental Period	Dirac Delta
Unit Step Function	$u(t) = \begin{cases} 0, & \text{if } t < 0 \\ 1, & \text{if } t \geq 0 \end{cases}$	$f(t) = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t)$	$\int_{-\infty}^{\infty} \delta(t) dt = 1$
Exponential Function	$f(t) = Ae^{at}$	$T_1 = \frac{1}{f_1} \qquad T_2 = \frac{1}{f_2}$	$\int_{-\infty}^{\infty} f(t)\delta(t - a)dt$
Sine Function	$f(t) = A\sin(\omega t + \phi)$	$T_0 = MT_1 = NT_2 \qquad \frac{M}{N} = \frac{f_1}{f_2}$	$= f(a)$
		$f_0 = \text{GCF}$	

Complex Numbers			Phasors		
Imaginary number $j = \sqrt{-1}$			$v(t) = V_m \cos(\omega t + \phi)$		
			$V = V_m \angle \phi = v_m e^{j\phi}$		
Form	z	z^*	Element	Law	Impedence z
Cartesian	$z = a + jb$	$z^* = a - jb$	Resistor R	$v(t) = Ri(t)$	$z = R$
Polar	$z = r \angle \theta$	$z^* = r \angle -\theta$	Inductor L	$v(t) = L \frac{di}{dt}$	$z = j\omega L$
Exponential	$z = re^{j\theta}$	$z^* = re^{-j\theta}$	Capacitor C	$i(t) = C \frac{dv}{dt}$	$z = \frac{1}{j\omega C}$
$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$					
$a = r \cdot \cos(\theta) \quad b = r \cdot \sin(\theta)$					

Euler's Identities	Shifted Cosines	$C_0 = a_0$
$e^{j\theta} = \cos(\theta) + j\sin(\theta)$	$y(t) = \frac{C_0}{2} + \sum_{n=1}^{\infty} [C_n \cos(n \cdot \omega_0 t + \theta_n)]$	$C_n = \sqrt{a_n^2 + b_n^2}$
$\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$		$\theta_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$
$\sin(\theta) = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$	Sum of Sines and Cosines	$\alpha_0 = \frac{a_0}{2}$
	$y(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$	$a_0 = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) dt$
		$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) \cos(n\omega_0 t) dt$
		$b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) \sin(n\omega_0 t) dt$
Roots of complex numbers	Complex Exponential	$\alpha_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) e^{-jn\omega_0 t} dt \quad \alpha_0 = \frac{a_0}{2}$
$x^n = a = a e^{j\theta}$	$y(t) = \sum_{n=-\infty}^{\infty} \alpha_n e^{jn\omega_0 t}$	$\alpha_n = \frac{a_n - jb_n}{2} \qquad \alpha_{-n} = \frac{a_n + jb_n}{2}$
$x = a ^{\frac{1}{n}} e^{j\left(\frac{\theta + 2\pi k}{n}\right)}$		
For n consecutive values of k		

Linear Algebra

$$\vec{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \vec{A}^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

Inner Product

$$\vec{a} = 1 \times 3 = [a_1 \quad a_2 \quad a_n]$$

$$\vec{b} = 3 \times 1 = \begin{bmatrix} b_1 \\ b_2 \\ b_n \end{bmatrix}$$

$$\vec{a}\vec{b} = a_1b_1 + a_2b_2 + a_nb_n$$

Scalar Multiplication

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Properties of Matrix Operations (Order Matters in Multiplication)

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$(\vec{A}\vec{B})^T = \vec{B}^T\vec{A}^T$$

$$\vec{A}\vec{B} \neq \vec{B}\vec{A}$$

$$\vec{C}(\vec{A} + \vec{B}) = \vec{C}\vec{A} + \vec{C}\vec{B}$$

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

$$\vec{A}(\vec{B}\vec{C}) = (\vec{A}\vec{B})\vec{C}$$

$$c(\vec{A}\vec{B}) = \vec{A}(c\vec{B}) = (c\vec{A})\vec{B}$$

Inverse

Matix $\vec{A}$ and $\vec{A}^{-1}$ are the inverse of each other if:

$$\vec{A}\vec{A}^{-1} = \vec{I} \quad \vec{A}^{-1}\vec{A} = \vec{I}$$

$$\vec{A} \cdot \vec{I} = \vec{A}$$

$$\vec{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Elementary row operations

1. Exchange 2 rows: $\vec{E}_{R1,R2}^x$
2. Multiply a row by a non-zero scalar: $\vec{E}_{c,R1}^M$
3. Multiply a row by a non-zero scalar and add it to another row: $\vec{E}_{c,R1,R2}^{M+}$

Determinant of a matrix

$$\vec{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow |\vec{A}| = ad - cb$$

$$\vec{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow$$

$$|\vec{A}| = a(ei - hf) - b(di - gf) + c(dh - ge)$$

Cramer's Rule

$$\vec{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \vec{A}^{-1} = \frac{\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}{|\vec{A}|}$$

Rotation and Translation

$$\vec{p} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

If $\theta > 0$ CCW, If $\theta < 0$ CW

If multiple translations in a row go backwards.

$$\vec{R}_0 \begin{bmatrix} \vec{p} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ 1 \end{bmatrix}$$

$$\vec{T}_{\Delta x, \Delta y} \begin{bmatrix} \vec{p} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \Delta x \\ 0 & 1 & \Delta y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ 1 \end{bmatrix}$$

Gaussian elimination

Given $\vec{A}\vec{x} = \vec{b}$, and asked to solve for $\vec{x}$

1. Put in form $[\vec{A}|\vec{b}]$
2. Use elementary row operations to convert $\vec{A} \rightarrow \vec{I}$ to the form $[\vec{I}|\vec{x}]$, Where $\vec{x}$ is the solution.

Over and Underdetermined Systems

n : # of variables m : # of equations

1. Overdetermined: $n < m$
 - a. No Solutions
 - b. Inconsistent
2. Critically determined: $n = m$
 - a. 1 Solution
 - b. Consistent
3. Undetermined: $n > m$
 - a. ∞ Solutions
 - b. Consistent

Laplace Transform

$$v(s) \triangleq \int_0^\infty v(t)e^{-st}dt = \mathcal{L}\{v(t)\}$$

$$v(t) \triangleq \frac{1}{2\pi j} \int_0^\infty v(s)e^{st}ds = \mathcal{L}^{-1}\{v(s)\}$$

Differential Equations

First Order

$$\frac{dv(t)}{dt} + av(t) = v_s(t)$$

Total Solution is:

$$v(t) = v_c(t) + v_p(t)$$

Solving for Particular:

1) Assume Solution: b & ω_0 is given

$$v_p(t) = \beta u(t)$$

$$v_p(t) = \beta e^{bt} u(t)$$

$$v_p(t) = \beta \cos(\omega_0 t + \phi) u(t)$$

2) Take derivatives and plug in to find unknowns.

Solving for Complementary:

1) Assume solution of form:

$$v_c(t) = Ce^{-at} u(t), a \text{ is given}$$

2) Combine with particular to make total solution and plug in initial conditions.

Second Order

$$\ddot{v}(t) + b\dot{v}(t) + cv(t) = v_s(t)$$

Solving for Particular: Same as first order.

Solving for Complementary:

1) Create characteristics equation:

$$\ddot{v}_c(t) + b\dot{v}_c(t) + cv_c(t) = 0$$

$$\lambda^2 + b\lambda + c = 0$$

$$\text{Damping Ratio: } \zeta = \frac{b}{2\sqrt{c}}$$

If $\zeta < 1 \rightarrow$ Underdamped

If $\zeta > 1 \rightarrow$ Overdamped

If $\zeta = 1 \rightarrow$ Critically damped

Underdamped: $\lambda = \alpha \pm j\beta$

$$v_c(t) = 2Ce^{\alpha t} \cos(\beta t + \phi) u(t)$$

Overdamped:

$$v_c(t) = (C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}) u(t)$$

Critically Damped:

$$v_c(t) = (C_1 e^{\lambda t} + t C_2 e^{\lambda t}) u(t)$$

Differential Equations

First Order

$$\frac{dv(t)}{dt} + av(t) = v_s(t)$$

Total Solution is $v(t) = v_c(t) + v_p(t)$

Where:

- v_c is the Complementary
- v_p is the Particular

$v_s(t)$ could come in many forms

1. Constant: $v_s(t) = Au(t)$
2. Exponential: $v_s(t) = Ae^{bt}u(t)$
3. Sinusoidal: $v_s(t) = A\sin(\omega_0 t + \phi)u(t)$

Solving for particular

1. Assume a solution
 - a. Constant: $v_p(t) = \beta u(t)$
 - b. Exponential: $v_p(t) = \beta e^{bt}u(t)$
 - c. Sinusoidal: $v_p(t) = \beta \cos(\omega_0 t + \phi)u(t)$
2. Take derivative and plug into to original equation to solve for unknown

Solving for Complementary

1. Assume solution of form $v_c(t) = Ce^{-at}u(t)$
2. Combine with particular to make total solution and plug in initial conditions.

Second Order

$$\ddot{v}(t) + B\dot{v}(t) + cv(t) = v_s(t)$$

Solving for particular

1. Assume a solution
 - a. Constant $v_p(t) = \beta u(t)$
 - b. Exponential $v_p(t) = \beta e^{bt}u(t)$
 - c. Sinusoidal $v_p(t) = \beta \cos(\omega_0 t + \phi)u(t)$
2. Take derivative and plug into to original equation to solve for unknown

Solving for Complementary

1. Create characteristics equation
$$\ddot{v}_c(t) + B\dot{v}_c(t) + cv_c(t) = 0$$

$$\lambda^2 + b\lambda + c = 0$$

Solve for roots and damping

$$\zeta = \frac{b}{2\sqrt{c}}$$

If $\zeta < 1 \rightarrow$ Underdamped

If $\zeta > 1 \rightarrow$ Overdamped

If $\zeta = 1 \rightarrow$ Critically damped

Underdamped: $v_c(t) = 2Ce^{\alpha t} \cos(\beta t + \phi) u(t)$, Where: $\alpha \pm j\beta$

Overdamped: $v_c(t) = (C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}) u(t)$

Critically Damped: $v_c(t) = (C_1 e^{\lambda t} + t C_2 e^{\lambda t}) u(t)$

2. Take derivative and plug into to original equation to solve for unknown

Euler's Method

$$\dot{v}(t) + f(v(t), t) = v_s(t)$$

$$\dot{v}(t) = \frac{v(t+h) - v(t)}{h}$$

Solving for $v(t+h)$

$$v(t+h) = v(t) - hf(v(t), t) + hv_s(t)$$

Let $k = 0, t_0 = 0$, Initial Conditions. These are given.

Plug in values into equation and increment k: $k = k + 1$

For Second Order

$$v''(t) + a(t)v'(t) + b(t)v(t) = v_s(t)$$

1. Define new variables (repeat for every higher order)

$$v_1(t) = v(t)$$

$$v_2(t) = \dot{v}(t) = \dot{v}_1(t)$$

2. Take derivatives and set up a system

$$\dot{v}_1(t) = v_2(t)$$

$$\dot{v}_2(t) = -b(t)v_1(t) - a(t)v_2(t) + v_s(t)$$

$$\begin{bmatrix} v_1(t+h) \\ v_2(t+h) \end{bmatrix} = \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} + h \begin{bmatrix} 0 & 1 \\ -b(t) & -a(t) \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} + h \begin{bmatrix} 0 \\ v_s(t) \end{bmatrix}$$

Laplace Transform

$$v(s) \triangleq \int_0^{\infty} v(t) e^{-st} dt = \mathcal{L}\{v(t)\}$$

$$v(t) \triangleq \frac{1}{2\pi j} \int_0^\infty v(s) e^{st} ds = \mathcal{L}^{-1}\{v(s)\} = v(t)$$

Laplace Transforms	Table 8.1
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$\sin(t)u(t)$	$\frac{1}{s^2+1}$
$\cos(t)u(t)$	$\frac{s}{s^2+1}$
$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$
$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$
$e^{at}u(t)$	$\frac{1}{s-a}$
$te^{at}u(t)$	$\frac{1}{(s-a)^2}$
$t^{n-1}e^{at}u(t)$	$\frac{(n-1)!}{(s-a)^n}$
$e^{at} \sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s-a)^2+\omega_0^2}$
$e^{at} \cos(\omega_0 t)u(t)$	$\frac{s-a}{(s-a)^2+\omega_0^2}$
$2 K e^{at}\cos(\omega_0 t + \angle K)u(t)$	$\frac{K}{s-(a+j\omega_0)} + \frac{K^*}{s-(a-j\omega_0)}$

$f(t)$	$F(s)$	Property
$av_1(t) + bv_2(t)$	$aV_1(s) + bV_2(s)$	Linearity
$v(t - t_0)$	$e^{-st_0}V(s)$	Time-shifting
$v(at)$	$\frac{1}{a}V\left(\frac{s}{a}\right)$	Time-scaling
$\frac{dv(t)}{dt}$	$sV(s) - v(0)$	First derivatives
$\frac{d^2v(t)}{dt^2}$	$s^2V(s) - sv(0) - \frac{dv(0)}{dt}$	Second derivatives
$\int_0^t v(\tau)d\tau$	$\frac{V(s)}{s}$	Integrals
$e^{-at}v(t)$	$V(s + a)$	Shifting in the s-domain

Partial Fractions

$$\mathcal{L}\{f^{(3)}(t)\} = s^3F(s) - s^2f(0^+) - sf'(0^+) - f''(0^+)$$

$$\frac{F(x)}{(x+a)(x+b)} = \frac{A}{x+a} + \frac{B}{x+b}$$

Where: A and B are residues.

In s Domain

1. Distinct, real roots:

$$\frac{F(s)}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b}$$

Where: $a \neq b$ and a, b are not complex

2. Repeated

$$\frac{F(s)}{(s+a)^n} = \frac{A_1}{s+a} + \frac{A_2}{(s+a)^2} + \dots + \frac{A_n}{(s+a)^n}$$

Where: a is not complex. The highest root is easy to solve, do it like normal. The lower exponents you will need to multiply the entire thing by the while denominator.

3. Complex roots

$$\frac{F(s)}{(s+a-jb)(s+a+jb)} = \frac{K}{s+a-jb} + \frac{K^*}{s+a+jb}$$

Shortcut trick

$$F(s) = \frac{G(s)}{(s-s_0)(s-s_1)} = \frac{A}{s-s_0} + \frac{B}{s-s_1}$$

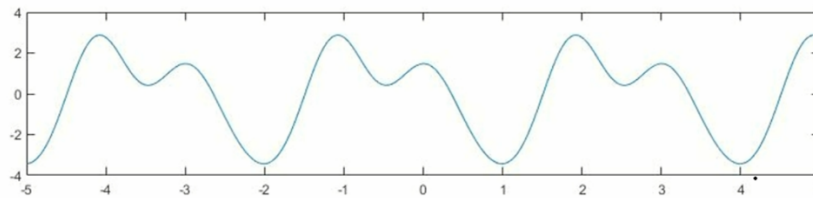
$$A = F(s)(s-s_0)|_{s=s_0}$$

$$B = F(s)(s-s_1)|_{s=s_1}$$

Fourier Series

The function has energy

We can represent any periodic signal as this:



$$y(t) = \frac{C_0}{2} + \sum_{n=1}^{\infty} [C_n \cos(n \cdot \omega_0 t + \theta_n)]$$

Where:

- C_n is the magnitude of the nth cosine (Energy),
- ω_0 is the fundamental frequency of the original signal.
- $C_0 = a_0$
- $C_n = \sqrt{a_n^2 + b_n^2}$
- $\theta_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$

$$y(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

Where:

- $\alpha_0 = \frac{a_0}{2}$

- $a_0 = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) dt$
- $a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) \cos(n\omega_0 t) dt$
- $b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) \sin(n\omega_0 t) dt$

Plugging Euler's and deriving, we arrive at:

$$y(t) = \sum_{n=-\infty}^{\infty} \alpha_n e^{jn\omega_0 t}$$

Where:

- $\alpha_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} y(t) e^{-jn\omega_0 t} dt$ $\alpha_0 = \frac{a_0}{2}$
- $\alpha_n = \frac{a_n - jb_n}{2}$
- $\alpha_{-n} = \frac{a_n + jb_n}{2}$

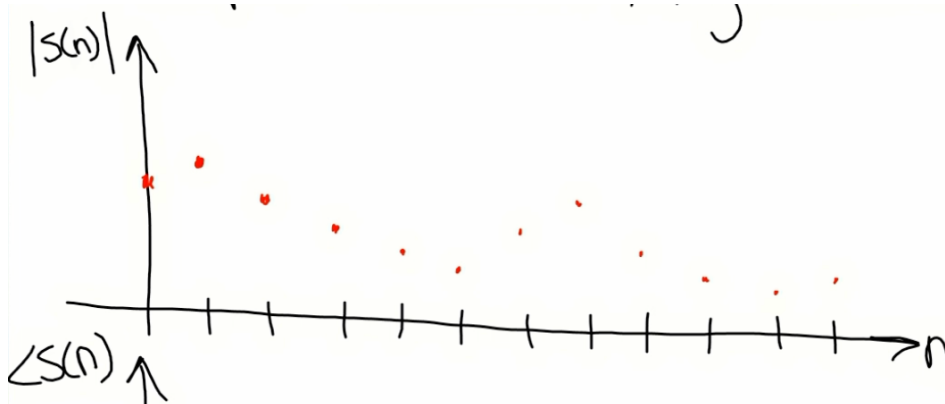
When asked to find the Fourier series of $s(t)$:

- 1) Determine which form it asks for
 - a. Shifted cosines (C_n, θ_n)
 - b. Sum of sines and cosines (a_n, b_n)
 - c. Complex exponential (α_n)
- 2) Solve for the variables ($C_n, \theta_n, a_n, b_n, \alpha_n$)
 - a. Given an equation for $s(t)$
 - i. Use algebra, trig identity, or Euler's
 - b. Given a graph of $s(t)$
 - i. Integrate to find (a_0, a_n, b_n) or (α_n)

The frequency spectrum of a signal

$\alpha_n = s(n) \rightarrow$ Discrete function

α_n is a complex function $\rightarrow$ Magnitude and phase graphs



Magnitude Spectrum
(Energy in frequencies)



Phase Spectrum
(Less important)